

C++ Programming Solutions

Question 1: Simple Calculator using switch

```
#include <iostream>
using namespace std;

int main() {
    float num1, num2;
    char op;

    cout << "Enter first number: ";
    cin >> num1;

    cout << "Enter second number: ";
    cin >> num2;

    cout << "Choose operation (+, -, *, /): ";
    cin >> op;

    switch(op) {
        case '+':
            cout << "Result: " << num1 + num2;
            break;
        case '-':
            cout << "Result: " << num1 - num2;
            break;
        case '*':
            cout << "Result: " << num1 * num2;
            break;
        case '/':
            if(num2 != 0) {
                cout << "Result: " << num1 / num2;
            } else {
                cout << "Error: Division by zero is not allowed.";
            }
            break;
        default:
            cout << "Invalid operator selected.";
    }

    return 0;
}
```

Question 2 Part A: Grade Calculator (if-else if-else)

```
#include <iostream>
using namespace std;
int main() {
```

```

int score;
cout << "Enter score (0-100): ";
cin >> score;

if(score >= 90 && score <= 100)
    cout << "Grade: A";
else if(score >= 80)
    cout << "Grade: B";
else if(score >= 70)
    cout << "Grade: C";
else if(score >= 60)
    cout << "Grade: D";
else if(score >= 0)
    cout << "Grade: F";
else
    cout << "Invalid score entered.";

return 0;
}

```

Question 2 Part B: Day of the Week Identifier (switch)

```

#include <iostream>
using namespace std;

int main() {
    int day;
    cout << "Enter number (1-7): ";
    cin >> day;

    switch(day) {
        case 1: cout << "Monday"; break;
        case 2: cout << "Tuesday"; break;
        case 3: cout << "Wednesday"; break;
        case 4: cout << "Thursday"; break;
        case 5: cout << "Friday"; break;
        case 6: cout << "Saturday"; break;
        case 7: cout << "Sunday"; break;
        default: cout << "Invalid input.";
    }

    return 0;
}

```

Question 3: Character Type Analyzer

```

#include <iostream>
using namespace std;

int main() {
    char ch;

```

```
cout << "Enter a character: ";
cin >> ch;

switch(ch) {
    case 'a': case 'e': case 'i': case 'o': case 'u':
    case 'A': case 'E': case 'I': case 'O': case 'U':
        cout << "\"" << ch << " is a vowel.";
        break;
    default:
        if((ch >= 'a' && ch <= 'z') || (ch >= 'A' && ch <= 'Z'))
            cout << "\"" << ch << " is a consonant.";
        else
            cout << "\"" << ch << " is not an alphabet character.";
}

return 0;
}
```